The Depth-Integrated Ocean Circulation
Munk's Theory

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Aim

The aim of this assignment is to study Munk's theory for the barotropic large-scale ocean circulation. To do this, implement some features from the previous assignments and model the large-scale circulation in a closed basin.

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Experiments and model setup

Model setup

- Let the system initially be in rest.
- Use solid boundaries.
- Program a wind forcing in the model, which is constant in x and sinusoidal in y (see the lecture notes).
- Use a β-plane.
- Program a bottom friction or viscosity in the form of a Laplace friction.

IMPORTANT REMARK : Do not forget to compute the barotropic streamfunction to compare your results with the analytical solution.

Experiment 1 - Sverdrup solution (frictionless)

- Run the model without any friction and use parameters that are suitable for the North Atlantic (D=3000m, $Lx = Ly = 7\cdot 10^6$, Nx=Ny=200, $f_0 = 7.27 \cdot 10^{-5}$, $\beta = 1.98 \cdot 10^{-11}$).
- Run the model with and without beta-plane
- Run the model with other values for τ_0.
- Run the model for the North Pacific, keep the same D but increase the domain size.

Experiment 2 - Munk's solution (viscosity)

- Run the model for the North Atlantic with a Laplace friction.
- Rerun the model using different values for the friction parameter, A.

Good luck!